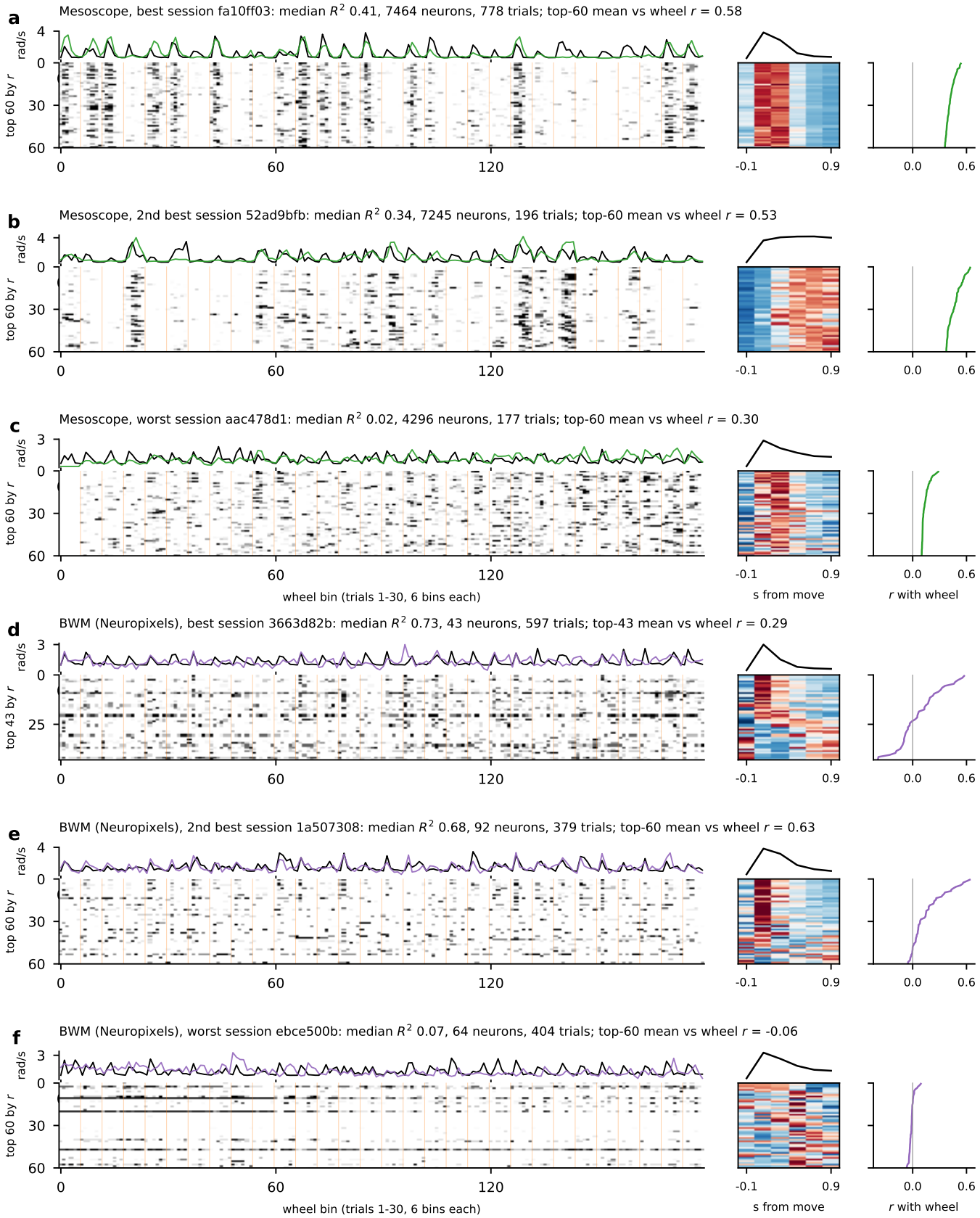


Figure 1: Neural correlates of wheel speed. Each row: wheel speed (top; rad/s) in the 6 wheel bins (0.2 s, -0.2 to 1 s around first movement) of the first 30 trials, concatenated (orange lines: trial boundaries; bins are not contiguous across trials; coloured: mean z-scored activity of the neurons below, rescaled), above the z-scored activity of the 60 neurons most positively correlated with wheel speed (sorted by r, highest on top). Neurons are ranked by r over all bins of the remaining trials, so the shown trials are held out from the selection. Middle: trial-averaged activity of the same 60 neurons per bin (z-scored per neuron, same order), with the mean wheel speed above. Right: their r with wheel speed over the remaining trials, same rows. Sessions at the best two and the worst of each modality by the session median R^2 over bins (all neurons per 0.4 mm bin, revised decoder with lagged features; sessions with $\geq$30 neurons). **a** Mesoscope session fa10ff03 (best of 149 sessions by the session median wheel-speed R^2): 95th percentile of r over neurons 0.20, maximum 0.54, 17% of neurons with $r > 0.1$; mean of the top 60 neurons vs wheel speed in the shown trials r = 0.58. **b** Mesoscope session 52ad9bfb (2nd best of 149 sessions by the session median wheel-speed R^2): 95th percentile of r over neurons 0.20, maximum 0.64, 18% of neurons with $r > 0.1$; mean of the top 60 neurons vs wheel speed in the shown trials r = 0.53. **c** Mesoscope session aac478d1 (worst of 149 sessions by the session median wheel-speed R^2): 95th percentile of r over neurons 0.07, maximum 0.29, 1% of neurons with $r > 0.1$; mean of the top 60 neurons vs wheel speed in the shown trials r = 0.30. **d** BWM (Neuropixels) session 3663d82b (best of 127 sessions by the session median wheel-speed R^2): 95th percentile of r over neurons 0.50, maximum 0.57, 42% of neurons with $r > 0.1$; mean of the top 43 neurons vs wheel speed in the shown trials r = 0.29. **e** BWM (Neuropixels) session 1a507308 (2nd best of 127 sessions by the session median wheel-speed R^2): 95th percentile of r over neurons 0.51, maximum 0.64, 30% of neurons with $r > 0.1$; mean of the top 60 neurons vs wheel speed in the shown trials r = 0.63. **f** BWM (Neuropixels) session ebce500b (worst of 127 sessions by the session median wheel-speed R^2): 95th percentile of r over neurons 0.05, maximum 0.09, 0% of neurons with $r > 0.1$; mean of the top 60 neurons vs wheel speed in the shown trials r = -0.06.